Calcurion Physics Worksheets

A-Level Physics

Topic: Circular Motion & SHM

MARKSCHEME

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1. Rate of change of angle; $\omega = 2\pi f$ **A1** [1]
2. $\omega = 4\pi$ rad/s; $v = 6.28$ m/s **M1A1** [3]
3. $a = v^2/r$ or $\omega^2 r$ **A1** [1]
4. $F = 4.5$ N **M1A1** [3]
5. Friction **A1** [1]
6. Acceleration proportional to displacement and towards equilibrium **A2** [2]
7. $a = -\omega^2 x$ **A1** [1]
8. $\omega = 4\pi$ rad/s **M1A1** [2]
9. $T = 2\pi\sqrt{m/k}$ **A1** [1]
10. $T = 0.314$ s **M1A1** [3]
11. $T = 2\pi\sqrt{l/g}$ **A1** [1]
12. $T = 2.0$ s **M1A1** [2]
13. Maximum displacement from equilibrium **A1** [1]
14. Equilibrium position **A1** [1]
15. Maximum displacement (amplitude) **A1** [1]
16. $v = \sqrt{GM/r} = 7.6 \times 10^3$ m/s **M1A1** [3]
17. For small θ, $a = -g\theta = -(g/l)x$; $\omega^2 = g/l$ **A4** [4]
18. $v_{\max} = \omega A = 0.785$ m/s; $a_{\max} = \omega^2 A = 12.3$ m/s^2 **M1A1** [4]
19. x and a in antiphase; v leads x by $\pi/2$ **A4** [4]
20. $E_k + E_p = \frac{1}{2}kA^2$ constant **A3** [3]
21. $A = 0.04$ m; $\omega = 5\pi$; $f = 2.5$ Hz **A3** [3]
22. $\omega = 10.5$; $KE_{\max} = \frac{1}{2}m\omega^2 A^2 = 0.044$ J **M1A1** [3]
23. g smaller; T larger **A2** [2]
24. $\tan\theta = v^2/(rg) = 39°$ **M1A1** [3]
25. Vary m; measure T; plot T^2 vs m; gradient $= 4\pi^2/k$ **A3** [3]